

Complete Prometheus + Grafana + Spring Boot + WSL2 Troubleshooting and Setup Guide

Overview

This document captures the full troubleshooting journey required to integrate a Spring Boot applicat.

Initial Architecture

Windows:

- IntelliJ IDEA
- Spring Boot
- Metrics endpoint on port 8090

WSL2:

- Docker Engine
- Prometheus
- Grafana

Problem 1: Docker Permission Denied

Error:
permission denied while trying to connect to docker.sock

Temporary fix:
sudo docker compose up -d

Permanent fix:
sudo usermod -aG docker \$USER
newgrp docker

Problem 2: Docker Network Creation Failure

Error:
failed to create network metrics_default: operation not supported

Investigation:
docker network create testnet

Kernel checks:
zgrep CONFIG_BRIDGE /proc/config.gz
zgrep CONFIG_VETH /proc/config.gz

Fix:
sudo sysctl -w net.ipv4.ip_forward=1

Persistence:
echo 'net.ipv4.ip_forward=1' | sudo tee /etc/sysctl.d/99-docker.conf
sudo sysctl --system

Problem 3: Prometheus Could Not Reach Spring Boot

Spring Boot metrics were accessible from Windows browser:

http://127.0.0.1:8090/metrics/prometheus

But inaccessible from WSL.

Spring Security Fix

Prometheus initially received:

401 Unauthorized
Solution:

```
.requestMatchers('/metrics/**').permitAll()
```

After restart:

```
HTTP 200 OK
```

Windows Host Discovery

```
WSL hostname:  
hostname -I
```

```
Windows adapter:  
Get-NetIPAddress -AddressFamily IPv4
```

```
Important IP:  
172.25.112.1
```

Port Proxy Solution

WSL could not access Windows localhost.

Created port proxy:

```
netsh interface portproxy add v4tov4  
listenaddress=172.25.112.1  
listenport=8091  
connectaddress=127.0.0.1  
connectport=8090
```

```
Verification:  
netsh interface portproxy show all
```

Firewall Rule

```
New-NetFirewallRule  
-DisplayName 'Spring Boot WSL Proxy 8091'  
-Direction Inbound  
-Action Allow  
-Protocol TCP  
-LocalPort 8091
```

Problem 4: Prometheus Target Timeout

Prometheus target remained DOWN:

```
context deadline exceeded
```

Container test:

```
docker exec -it prometheus sh
```

```
wget http://172.25.112.1:8091/metrics/prometheus
```

Timeout occurred.

Root Cause

Docker bridge networking inside WSL was unable to reliably reach the Windows portproxy endpoint.

Proof:

```
docker run --rm --network host alpine sh -c  
'wget -qO- http://172.25.112.1:8091/metrics/prometheus | head'
```

Succeeded immediately.

Prometheus Host Networking Fix

```
docker-compose.yml
```

```
prometheus:
```

```
image: prom/prometheus:latest
network_mode: host
```

Prometheus Configuration

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: prometheus
    static_configs:
      - targets:
        - localhost:9090

  - job_name: spring-boot
    metrics_path: /metrics/prometheus
    static_configs:
      - targets:
        - 172.25.112.1:8091
```

Verification

```
Prometheus:

http://localhost:9090/targets

Expected:

prometheus UP
spring-boot UP
```

Grafana Setup

```
Grafana URL:

http://localhost:3000

Default login:
admin/admin
```

Grafana Datasource

```
Prometheus datasource URL:

http://172.25.124.34:9090

NOT localhost:9090

Reason:
Grafana runs in a container.
```

Useful Metrics

```
sum(jvm_memory_used_bytes{area='heap'})

process_cpu_usage * 100

jvm_threads_live_threads

hikaricp_connections_active

rate(http_server_requests_seconds_count[1m])
```

Custom Business Metrics

```
Counter.builder('loopin.user.registration')
Counter.builder('loopin.post.created')
Counter.builder('loopin.like.created')

Then increment counters and monitor through Grafana.
```

Final Working Endpoints

Spring Metrics:
<http://172.25.112.1:8091/metrics/prometheus>

Prometheus:
<http://localhost:9090>

Grafana:
<http://localhost:3000>

Grafana Datasource:
<http://172.25.124.34:9090>